

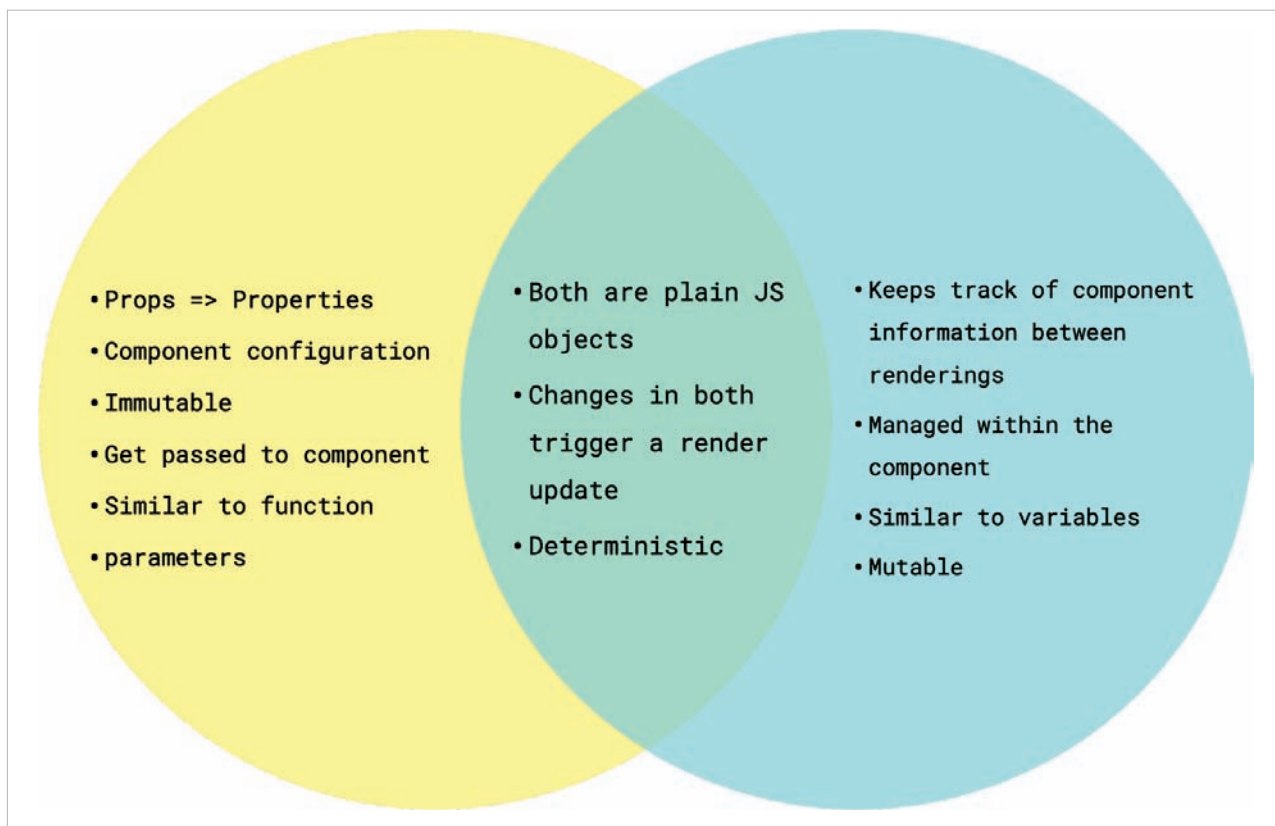


Figure 1: Props and state

Now that we know the advantages of using React, let's look at its concepts. The two popular ones are props and state. ReactJS being a component based library, the user interface is divided into small reusable pieces called components. These components may need to send data to each other in some scenarios, and this passing of data between the components is done using props. As you may already have guessed, props stand for properties and are used for passing data from one component to another, usually between the parent and the child. Props are always read-only, that is, you cannot change them. So once the child component receives the data from the parent component, the child component can only use it and not change it at all.

The next concept is state. State is a built-in React object that is used to contain data or information about the component. The component state can change over time, and whenever it changes, the component re-renders. The change in state can occur as a response to user action or system-generated events, and these changes determine the behaviour of the component and how it will render on the UI. The state object is initialised in the constructor and can store multiple properties.

We use *this.setState* to change the value of the state object and will look at how to do that in a short time.

Stateful component

```
function MyApp(props) {
  return <h1>Hello, {props.name}</h1>
}
```

Using ES6 class

```
class MyApp extends React.Component {
  render() {
    return <h1>Hello, {props.name}</h1>;
  }
}
```

React basically has two types of components:

- Simple component
 - Stateful component
- The simple component does not have a state, although it can get the values through props from the parent component. You can see the first part of the code above shows a functional component that takes in the *name* from props.

And in the second part of the code, you can see that a similar result is achieved, but this time we use a class. All the class components extend the React component and have a